DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT25-425 P-II
COMPUTER NETWORKS
(MORNING/EVENING)

LAB#08

TASK: IMPLEMENTATION OF VLAN USING PACKET TRACER

Name: _____ Roll No:_____

Score: _____ Signature of the Lab Tutor _____ Date: _____

Objectives:

1. To perform basic configuration tasks on a switch
2. Create VLAN and assign switch ports to a VLAN
3. Test and verify the configuration

Description

Virtual LAN:

A Virtual Local Area Network (VLAN) is a logical grouping of network users and resources connected to administratively defined ports on a switch. By creating VLANs, you are able to create smaller broadcast domains within a switch by assigning different ports in the switch to different subnetworks.

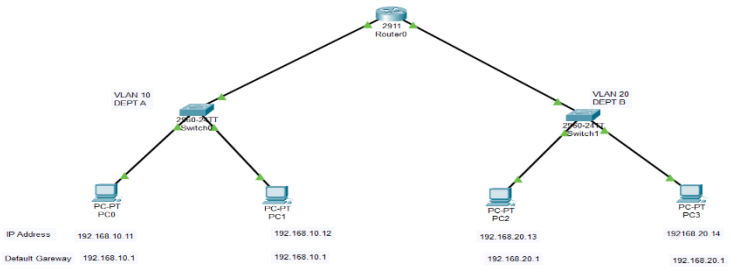
Key Aspects of VLANs

- **Improved Security:** Sensitive traffic (e.g., Finance) is isolated from general traffic (e.g., Guests).
- **Enhanced Performance:** Limits broadcast domains, reducing unnecessary network load.
- **Flexibility:** Devices can be moved logically between networks without changing physical cables.
- **Cost Efficiency:** Eliminates the need for multiple physical switches for different networks.
- **Management:** Simplifies administration by grouping users by department or function.

Procedure

To implement this practical following network topology is required to be configured using the VLAN commands. After configuring the given network, a packet should be ping from any one machine to another.

Topology



Task 1: Prepare the Network

Cable a network that is similar to the one in the topology diagram.
You can use any current switch in your lab as long as it has the required interfaces shown in the topology.

Task 2: Create VLANs on switch Switch0 and Switch1.

There are Two VLANS configured for this lab: VLAN 10 (Dept_A); VLAN 20 (Deapt_B). After you create the VLAN, you will be in vlan configuration mode, where you can assign a name to the VLAN with the name vlan name command.

Switch 0

```
Switch>enable
Switch#config t
Switch(config)#vlan 10
Switch(config-vlan)#name Dept_A
Switch(config-vlan)#exit
Switch(config)#interface range fa0/1 - 24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
```

Verify that the VLANs have been created on Switch0 and Switch1.
Use the show vlan brief command to verify that the VLANs have been created.

Switch#show vlan brief

Switch 1

```
Switch>enable
Switch#config t
Switch(config)#vlan 20
Switch(config-vlan)#name Dept_B
Switch(config-vlan)#exit
```

```
Switch(config)#interface range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
```

Verify that the VLANs have been created on Switch0 and Switch1.
Use the show vlan brief command to verify that the VLANs have been created.

```
Switch#show vlan brief
```

Task 3: Configure and Activate Ethernet Interfaces on Router0

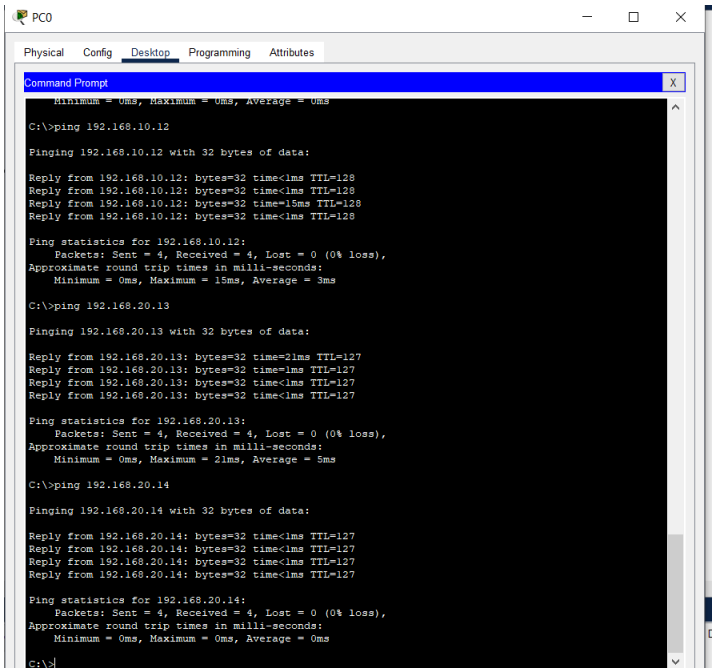
Router 0

```
Router>enable
Router#config t
Router(config)#interface gig0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#description Dept_A
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
```

```
Router(config-if)#end
```

```
Router>enable
Router#config t
Router(config)#interface gig0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#description Dept_B
Router(config-if)#no shutdown
```

Verify the interconnectivity and communication between the two vlans.



Review Questions:

1. What is a VLAN?

_____

_____

2. What are the advantages of VLAN?

_____

_____

3. How do you verify VLAN configuration on a Cisco switch?

_____

_____

4. How do you troubleshoot VLAN issues on a network?

_____

_____

Note: Submit the handout before the next Lab